Requirement Document: N8N Hands-On Intermediate Course

Topic 3: Data Validation, Cleaning & Duplicate Prevention

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1. Document Control

Field	Value
Document Title	Topic 3: Data Validation, Cleaning & Duplicate Prevention - Requirement File
Course	N8N - Hands On Intermediate Course (Level 2)
Target Audience	Automation Builders, System Integrators, QA Engineers
Version	1.0
Status	Approved
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Prepared By	Lead Automation Architect

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2. Learning Concepts & Objectives

2.1 Course Objective

Build an enterprise lead intake sanitization and deduplication pipeline in n8n. The workflow ingests raw, messy marketing leads, normalizes phone numbers into E.164 standard format, validates email structures using RFC regex, filters out disposable/temporary email domains, removes intra-batch

duplicates using set-based hash indexing, dispatches clean unique leads to a mock CRM, and routes bad/duplicate leads to segregated quarantine logs.

2.2 Learning Scope

- **Marketing Lead Ingestion:** Emitting mock batches of dirty marketing leads with common real-world flaws (missing country codes, malformed email syntax, disposable domains, and intra-batch duplicates).
- **Phone Sanitization & E.164 Formatting:** Stripping non-numeric characters and standardizing telephone numbers to the ITU-T E.164 international standard (`+1` for 10-digit US numbers).
- **Multi-Layer Quality Screening:** Applying RFC 5322 regex checks for email syntax, blacklisting temporary email domains (`mailinator.com`, `tempmail.com`, `10minutemail.com`, `guerrillamail.com`, `trashmail.com`), and enforcing minimum name length constraints.
- **Intra-Batch Deduplication:** Leveraging JavaScript `Set` data structures to detect duplicate submissions across email and phone fingerprints in a single execution batch.
- **Segregated Quarantine Architecture:** Routing syntax/spam rejections to `Log_QuarantineInvalid` and duplicate records to `Log_QuarantineDuplicate` for auditing and compliance.
- **CRM Synchronization & Batch Analytics:** Posting clean leads to a CRM ingestion endpoint and compiling execution batch metrics.

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3. Workflow Architecture

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graph TD

    Trigger[Node 1: Manual_Trigger] --> RawLeads[Node 2: Code_GenerateRawLeads<br/>Simulates 5 Dirty Leads]

    RawLeads --> Sanitize[Node 3: Code_SanitizeAndFormat<br/>E.164 Phone & Lowercase Email]

    Sanitize --> Validate[Node 4: Code_ValidateLeadQuality<br/>RFC Regex & Disposable Domain Check]


    Validate --> QualityIF{Node 5: IF_IsQualityLead<br/>isValid == true?}


    QualityIF -- "False (Invalid)" --> QuarInvalid[Node 7: Log_QuarantineInvalid<br/>SYNTAX_OR_SPAM_VIOLATION]

    QualityIF -- "True (Valid)" --> Dedup[Node 6: Code_DeduplicateLeads<br/>Set Hash Deduplication]


    Dedup --> UniqueIF{Node 8: IF_IsUniqueLead<br/>isDuplicate == false?}


    UniqueIF -- "False (Duplicate)" --> QuarDup[Node 10: Log_QuarantineDuplicate<br/>DUPLICATE_RECORD_SUPPRESSED]

    UniqueIF -- "True (Clean & Unique)" --> PostCRM[Node 9: HTTP_PostCleanLeadToCRM<br/>POST Clean Lead to CRM]


    PostCRM --> Metrics[Node 11: Code_PipelineMetrics<br/>LEAD_CLEANSING_BATCH_COMPLETE]
```

```
    style Trigger fill:#f0f9ff,stroke:#0284c7,stroke-width:2px
    style QualityIF fill:#fefce8,stroke:#ca8a04,stroke-width:2px
    style UniqueIF fill:#fefce8,stroke:#ca8a04,stroke-width:2px
    style PostCRM fill:#f0fdf4,stroke:#16a34a,stroke-width:2px
    style QuarInvalid fill:#fef2f2,stroke:#dc2626,stroke-width:2px
    style QuarDup fill:#fef2f2,stroke:#dc2626,stroke-width:2px
```

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4. Functional Requirements

Requirement ID	Section	Requirement Description	Priority
REQ-001	Lead Ingestion	The workflow shall simulate or ingest raw marketing submissions containing name, email, phone, company, and notes.	High
REQ-002	E.164 Phone Normalization	Telephone numbers shall be stripped of formatting characters (dashes, spaces, parentheses) and standardized to E.164 international notation.	High
REQ-003	RFC Email Validation	Incoming emails shall be validated against the RFC 5322 regex pattern; malformed addresses shall be flagged immediately.	High
REQ-004	Disposable Domain Filtering	Domains matching known throwaway email services (e.g. `mailinator.com`, `tempmail.com`) shall be detected and rejected.	High
REQ-005	Invalid Lead Quarantine	Defective leads failing syntax or domain checks shall route to a dedicated quarantine handler and not proceed to deduplication or CRM sync.	High
REQ-006	In-Memory Deduplication	Validated leads shall be compared across the batch using email and phone sets to detect repeated submissions.	High
REQ-007	Duplicate Lead Quarantine	Detected duplicate records shall route to a duplicate quarantine logger capturing the specific collision reason.	High
REQ-008	CRM Ingestion & Metrics	Exactly clean, unique leads shall be dispatched to the CRM endpoint, and overall batch metrics shall be compiled.	High

5. Technical Specifications & Node Blueprint

Node ID	Node Name	Node Type	Critical Role & Configuration
N3-01	`Manual_Trigger`	Manual Trigger	On-demand test runner initiating the cleansing pipeline.
N3-02	`Code_GenerateRawLeads`	Code (JavaScript)	Generates 5 mock leads (1 valid US, 1 invalid email, 1 disposable email, 1 duplicate, 1 valid UK).
N3-03	`Code_SanitizeAndFormat`	Code (JavaScript)	Lowercases email, standardizes phone to E.164 (`+1...`), and trims text strings.
N3-04	`Code_ValidateLeadQuality`	Code (JavaScript)	Evaluates RFC email regex, disposable blacklist, E.164 regex, and minimum name length.
N3-05	`IF_IsQualityLead`	IF Node	Routes leads based on `{{ $json.isValid }} == true`.
N3-06	`Log_QuarantineInvalid`	Code (JavaScript)	Logs syntax and spam violations (`SYNTAX_OR_SPAM_VIOLATION`) with timestamps.
N3-07	`Code_DeduplicateLeads`	Code (JavaScript)	Performs in-batch hash indexing via `Set()` on email and phone keys.
N3-08	`IF_IsUniqueLead`	IF Node	Routes leads based on `{{ $json.isDuplicate }} == false`.
N3-09	`HTTP_PostCleanLeadToCRM`	HTTP Request	Dispatches verified unique leads to CRM mock endpoint (`POST https://httpbin.org/post`).
N3-10	`Log_QuarantineDuplicate`	Code (JavaScript)	Logs duplicate suppression events (`DUPLICATE_RECORD_SUPPRESSED`).
N3-11	`Code_PipelineMetrics`	Code (JavaScript)	Aggregates synchronized CRM leads and emits completion summary metrics.

6. Test Cases & Verification Guide

ID	Test Case	Preconditions	Steps	Expected Result	Priority
TC-001	End-to-End Pipeline Execution	n8n canvas ready.	Click **Execute workflow** from `Manual_Trigger`.	Workflow completes with all nodes green; exactly 2 leads reach CRM, 2 quarantined as invalid, 1 quarantined as duplicate.	High
TC-002	Syntax Error Quarantine	Mock lead `LD-002` emitted.	Check `Log_QuarantineInvalid` output.	`LD-002` quarantined with reason `INVALID_EMAIL_SYNTAX`.	High
TC-003	Disposable Domain Quarantine	Mock lead `LD-003` emitted.	Check `Log_QuarantineInvalid` output.	`LD-003` quarantined with reason `DISPOSABLE_EMAIL_DOMAIN_BLOCKED`.	High
TC-004	Intra-Batch Duplicate Detection	Mock lead `LD-004` emitted.	Check `Log_QuarantineDuplicate` output.	`LD-004` quarantined with reason `Duplicate Email: sarah.connor@skynet-resist.org`.	High
TC-005	Clean CRM Synchronization	Mock leads `LD-001` and `LD-005` emitted.	Check `HTTP_PostCleanLeadToCRM` output.	Both leads received with E.164 phone numbers (`+15552345678` and `+442079460919`) and 200/201 response.	High
TC-006	Final Pipeline Metrics Verification	Sync completed.	Check `Code_PipelineMetrics` output.	Emits `LEAD_CLEANSING_BATCH_COMPLETE` with `leadsSentToCRM: 2`.	High

7. Assumptions, Dependencies, and Risks

7.1 Assumptions

- ASM-001: Input marketing payloads provide string fields for name, email, and phone.
- ASM-002: US 10-digit phone numbers require the `+1` prefix for international E.164 compliance.

7.2 Dependencies

- DEP-001: n8n instance version 1.0.0 or higher.
- DEP-002: External CRM endpoint (`https://httpbin.org/post` or mock URL) accessible via HTTPS.

7.3 Risks & Mitigations

- **RSK-001 (Shared Switchboard Numbers):** Multiple leads sharing a generic office landline could collide in phone deduplication.
  Mitigation: Prioritize email uniqueness first, and use compound identity verification.
- **RSK-002 (Disposable Domain Drift):** New disposable email domains emerge continuously.
  Mitigation: Maintain an external or centrally updated disposable domain blocklist.